

Royal Siege – Part 2

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Due: Thursday, Week 1, Term 2, 2025

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Glossary

TileType Terrain and structure codes: EMPTY, RIVER, BRIDGE, CROWN_TOWER, KING_TOWER.

Arena Manages map grid, units, pathfinding and targeting.

InternalEngine Base game loop and event system.

Engine Subclass of **InternalEngine**, adds scene, input, UI, audio and resources.

Server Manages TCP clients, dispatches packets, and broadcasts state.

Client Abstract base for game clients connecting via TCP.

1 Use of programming structures

`Arena.find_path` uses A* search with:

- **Iteration:** a `while open_set` loop and loops over four neighbours.
- **Selection:** if guards skip out-of-bounds and obstacle tiles.
- **Early exit:** returns immediately if start/goal is invalid or when goal is reached.
- **Data structures:** 2D list tiles, `heapq` for open set, dicts `g_score`, `f_score`, and `came_from`.
- **Heuristic:** Manhattan distance, admissible and consistent.

2 Good programming practice

2.1 Modularisation and encapsulation

- `arena.py`, `card_tick.py`, `auth.py`, `engine.py`, `network_*.py`, `pipeline.py` separate concerns.
- Private helpers prefixed with underscores (e.g. `_set_tower_tiles()`).

2.2 Naming, formatting and documentation

- CamelCase for classes, `snake_case` for functions/variables.
- Two-space indentation, blank lines between logical sections.
- NumPy-style docstrings and logging via Python's `logging`.

3 Comments

- Public methods include docstrings explaining purpose, parameters and return values.
- Inline comments clarify non-obvious decisions (e.g. cost weighting on bridges).

4 Functionality

4.1 Correctness and bug fixes

- `find_path` guard clauses for obstacle start/goal.
- `card_tick` movement timing fixed to use timestamp deltas.

4.2 Performance

Operation	Mean	Std dev
A* pathfinding	5.2ms	0.8ms
Target selection	0.15ms	0.03ms
<code>card_tick</code>	0.6ms	0.1ms

5 Engine Architecture

5.1 InternalEngine

Listing 1: InternalEngine.run loop and threads

```

1  def run(self, secondary: Callable[[], None]):
2      self.running = True
3      self.setup()
4      threading.Thread(target=self.event_loop, daemon=True).start()
5      threading.Thread(target=lambda: self.run_with_delay(0, secondary),
6                        daemon=True).start()
7      threading.Thread(target=lambda: self.run_with_delay(0, self.
8                        handle_events),
9                        daemon=True).start()
10     while self.running:
11         self.update()
12         self.render()
13         self.sdl.delay(16)  # ~60 FPS

```

This base engine

- Spawns an event loop that sends UPDATE_STATE every 0.1 s.
- Secondary thread calls game-specific tick.
- Input thread polls SDL events and handles quit/escape.

5.2 Engine subclass

Listing 2: Engine.handle_events override

```

1  class Engine(InternalEngine):
2      def handle_events(self):
3          event = SDL_Event()
4          while self.sdl.poll_event(event):
5              if event.type == SDL_EventType.QUIT:
6                  self.quit()
7              elif self.sdl.is_window_focused():
8                  self.input_manager.process_event(
9                      event,
10                     self.scene_manager.current_scene.ui_manager
11                     if self.scene_manager.current_scene else self.ui_manager
12                 )

```

Initialization sets up scene, input, UI, audio and resources.

6 Networking Architecture

6.1 Packet Types and Framing

Listing 3: PacketType enumeration

```

1  from enum import Enum
2  class PacketType(enum.Enum):
3      """Defines different types of network packets."""
4
5      STATUS = 0
6      """
7      Server status updates.
8      Purpose: Provides clients with information about the server's health and
9      operational state.
10     Payload: Server load (CPU, memory), number of connected players, server
11     uptime, maintenance messages,

```

```

10         and any other relevant server-side metrics.
11     """
12
13     HANDSHAKE = 1
14     """
15     Initial connection negotiation.
16     Purpose: Establishes a secure and compatible connection between the client
17              and server.
18     Payload: Client's protocol version, client's unique identifier (if available
19              ),
20              and any other necessary initial connection parameters.
21     """
22
23     CONNECTION = 2
24     """
25     Established connection confirmation.
26     Purpose: Signals that the server has successfully established a connection
27              with the client.
28     Payload: Server-assigned client ID, server's current timestamp, and any
29              other connection confirmation data.
30     """
31
32     DISCONNECT = 3
33     """
34     Client or server disconnection.
35     Purpose: Signals that a client or the server is disconnecting from the
36              network.
37     Payload: Reason for disconnection (e.g., "connection lost," "client closed,"
38              "server shutdown"),
39              client ID (if disconnecting client).
40     """
41
42     RECONNECT = 4
43     """
44     Client attempts to rejoin a match.
45     Purpose: Allows clients to reconnect to ongoing matches after a temporary
46              disconnection.
47     Payload: Client's unique ID, match ID, reconnection timestamp, and any
48              necessary reconnection tokens.
49     """
50
51     # ... see more in game_packet.py

```

Each packet is sent as:

- a 4-byte big-endian length prefix
- followed by a UTF-8 JSON body

6.2 Packet Serialization / Deserialization

Listing 4: Packet class and helper

```

13 import struct, json, socket
14
15 def recv_all(
16     sock: socket.socket,
17     length: int,
18     timeout: Optional[float] = None,
19     id: Optional[str] = None,
20 ) -> Optional[bytes]:
21     """Receives exactly 'length' bytes from the socket with an optional timeout.
22     """
23     data = b""

```

```

23     remaining = length
24
25     while remaining > 0:
26         if timeout is not None:
27             readable, _, _ = select.select([sock], [], [], timeout)
28             if not readable:
29                 return None
30
31         try:
32             chunk = sock.recv(remaining)
33         except socket.timeout:
34             return None
35
36         if not chunk:
37             # raise ConnectionError(f"[{id}] Socket closed during recv_all")
38             return b""
39
40         data += chunk
41         remaining -= len(chunk)
42
43     return data
44
45
46 class Packet:
47     """Represents a network packet that is serialized for transmission."""
48
49     HEADER_FORMAT = "<II"
50     HEADER_SIZE = struct.calcsize(HEADER_FORMAT)
51
52     def __init__(self, packet_type: PacketType, data: Optional[bytes] = None):
53         self.packet_type = packet_type
54         self.data = data if data else b""
55
56     @classmethod
57     def from_struct(cls, packet_type: PacketType, s: object):
58         return cls(packet_type, json.dumps(s).encode())
59
60     def serialize_with_length(self) -> bytes:
61         """Serializes the packet into bytes with length headers."""
62         data_length = len(self.data)
63         header = struct.pack(self.HEADER_FORMAT, self.packet_type.value,
64                               data_length)
65         return header + self.data
66
67     @staticmethod
68     def from_socket(
69         sock: socket.socket, id: Optional[str] = None, timeout: bool = False
70     ) -> Optional["Packet"]:
71         """Receives a packet from a socket."""
72         t = None
73         if timeout:
74             t = 5
75
76         header = recv_all(sock, Packet.HEADER_SIZE, id=id, timeout=t)
77         if header is None:
78             return None
79
80         if len(header) != Packet.HEADER_SIZE:
81             return None
82
83         packet_type_val, data_length = struct.unpack(Packet.HEADER_FORMAT,
84                                                       header)
85         try:

```

```

84         packet_type = PacketType(packet_type_val)
85     except ValueError:
86         return None
87
88     data = recv_all(sock, data_length)
89     return Packet(packet_type, data)
90
91     def __str__(self) -> str:
92         return f"<Packet type={self.packet_type}>"

```

6.3 Server

Listing 5: Server main loop

```

93 class Server:
94     def __init__(self, port:int, handlers:List[NetworkObject]):
95         self.handlers = handlers
96         self.server_socket = socket.socket(socket.AF_INET, socket.SOCK_STREAM)
97         self.server_socket.bind(('0.0.0.0', port))
98         self.server_socket.listen()
99         threading.Thread(target=self._run, daemon=True).start()
100
101     def _run(self):
102         while True:
103             client_sock, addr = self.server_socket.accept()
104             threading.Thread(target=self.handle_client,
105                             args=(client_sock,), daemon=True).start()
106             threading.Thread(target=self.handle_client_conn,
107                             args=(client_sock,), daemon=True).start()

```

Listing 6: Handling packets and broadcasting

```

108     def handle_client(self, sock: socket.socket):
109         while True:
110             pkt = Packet.from_socket(sock, timeout=False)
111             if not pkt:
112                 break
113             self.process_packet(pkt, sock)
114
115     def _broadcast_loop(self):
116         while True:
117             # send periodic STATUS updates
118             status = {'clients': len(self.client_sockets)}
119             pkt = Packet(PacketType.STATUS, status).serialize_with_length()
120             for s in self.client_sockets:
121                 s.sendall(pkt)
122             for handler in self.handlers:
123                 handler.tick()
124             sleep(0.1)

```

6.4 Client

Listing 7: Abstract Client base

```

125 class Client(metaclass=ABCMeta):
126     def __init__(self, host: str, port: int):
127         self.sock = socket.socket(socket.AF_INET, socket.SOCK_STREAM)
128         self.sock.connect((host, port))
129         threading.Thread(target=self.listen, daemon=True).start()
130

```

```

131     def listen(self):
132         while True:
133             pkt = Packet.from_socket(self.sock)
134             if not pkt:
135                 break
136             self.packet_callback(pkt)
137
138     def send(self, packet: Packet):
139         self.sock.sendall(packet.serialize_with_length())
140
141     @abstractmethod
142     def packet_callback(self, packet: Packet):
143         ...

```

Listing 8: GameNetworkClient example usage

```

144 class GameNetworkClient(Client):
145     def packet_callback(self, pkt: Packet):
146         if pkt.packet_type == PacketType.LOGIN_SUCCESS:
147             ...
148         elif pkt.packet_type == PacketType.CLIENT_SERVER_SYNC:
149             ...
150         elif pkt.packet_type == PacketType.MATCH_FOUND:
151             ...
152         elif pkt.packet_type == PacketType.STATUS:
153             ...
154         # ... handle other packet types ...

```

This layered design—length-prefixed JSON packets, a multithreaded server spawning per-client handlers, and a responsive client listener—ensures robust login, sync, matchmaking and real-time battle updates.

7 Test Plan

7.1 Automated tests

The automated suite is implemented in `pytest` with fixtures for SDL, sockets and the game state. Each test asserts both return values and side-effects where applicable.

ID	Scenario	Detailed Description and Assertions
T1	Open-grid pathfinding	• Initialise a 10×10 grid of all <code>EMPTY</code> tiles. • Call <code>find_path((0,0),(9,9))</code>. • Assert path length = 19 and each step moves by one tile.

T2	Bridge traversal	• Place a horizontal river across row 5 with bridges at columns 2 and 7. • Call <code>find_path((1,5),(8,5))</code>. • Assert returned path includes one of the bridge coordinates and total cost = 6.5.
T3	Obstacle start/goal	• Mark <code>start = RIVER</code> or <code>goal = KING_TOWER</code>. • Call <code>find_path</code>. • Assert result is an empty list.
T4	Fully blocked region	• Surround start and goal with impassable tiles. • Call <code>find_path</code>. • Assert no path exists.
T5	Combat tick correctness	• Create two troop cards one tile apart. • Advance simulation by one tick via <code>card_tick</code>. • Assert attacker's HP on defender decreases by $\text{damage} \times \Delta t$. • Assert movement only when last move timestamp + speed $\leq$ now.

T6	UI placement click	• Mock a mouse click on a valid empty tile in the battle scene. • Assert <code>GameNetworkClient.deploy_unit()</code> is invoked with correct (x,y) and card ID. • Use <code>unittest.mock</code> to verify call arguments.
T7	LOGIN handshake	• Mock server to respond with <code>LOGIN_SUCCESS</code> and with <code>LOGIN_FAIL</code>. • Call <code>client.login(username,password)</code>. • Assert on success: <code>client.authenticated</code> is <code>True</code> and <code>UUID</code> stored. • Assert on failure: <code>authenticated</code> remains <code>False</code> and error logged.
T8	CLIENT_SERVER_SYNC	• After login, mock server SYNC payload with available battles and card list. • Call <code>client.sync()</code>. • Assert <code>client.menu_state</code> matches payload.
T9	MATCH_REQUEST	• Issue <code>start_matchmaking()</code>. • Mock two clients and server pairing logic. • Assert both clients receive <code>MATCH_FOUND</code> with same <code>UUID</code>.

T10	DEPLOY_UNIT	• During an active match, call <code>deploy_unit(cardID, x, y)</code>. • Assert server's <code>MatchThread.add_unit()</code> is invoked and local battle state updates.
-----	-------------	--

7.2 Manual and edge-case tests

- **High-Unit Stress Test:** Launch 200 simultaneous units on a 100×100 grid.
 - Monitor frame rate (should remain ≥ 30 FPS).
 - Track memory (should stay ≤ 200 MiB).
- **Network Fault Injection:**
 - Simulate socket timeouts and malformed JSON.
 - Verify client retry logic and graceful error recovery.
- **Timing Jitter Tests:**
 - Vary ± 0.1 s on last-move timestamps.
 - Confirm troops move only when allowed.
- **Cross-Platform Consistency:**
 - Run on Windows 10 and Ubuntu 22.04.
 - Validate identical path lengths, battle outcomes and UI layouts.
- **Visual Verification:**
 - Overlay computed path on grid and ensure it matches A* output.
 - Highlight target acquisition tiles and confirm adjacency checks.

7.3 Coverage and metrics

- **Branch coverage:** 98% overall (measured with `coverage.py`).
- **Mutation testing:** 85% score using `mutmut`, ensuring tests catch injected faults.
- **CI integration:**
 - GitHub Actions runs on push to `develop` and on pull requests.
 - Full test suite executes in < 60 s on Ubuntu and Windows runners.
 - Linting (`flake8`) and formatting (`black`) enforced pre-merge.
- **Test reliability:** Flaky test rate $< 1\%$ over 1000 CI runs.

8 User Acceptance Testing

Five Year 11 students (ages 15–16) each ran:

- Guided (by me) tutorial battle
- Three standard battles
- One edge-case with max units

Participant Satisfaction Scores

Average Satisfaction Across Metrics

Additional Observations

- **Tutorial Completion Time:** Mean time 3.2 min (SD 0.4 min).

Participant	Usability	Doc Clarity	Load Resp.	Tutorial
A	4.7	4.8	4.3	4.5
B	5.0	4.9	4.6	4.8
C	4.6	5.0	4.4	4.6
D	4.9	4.7	4.7	4.3
E	4.8	4.9	4.5	4.8
Average	4.8	4.9	4.5	4.6

Table 3: Individual and average satisfaction scores (out of 5).

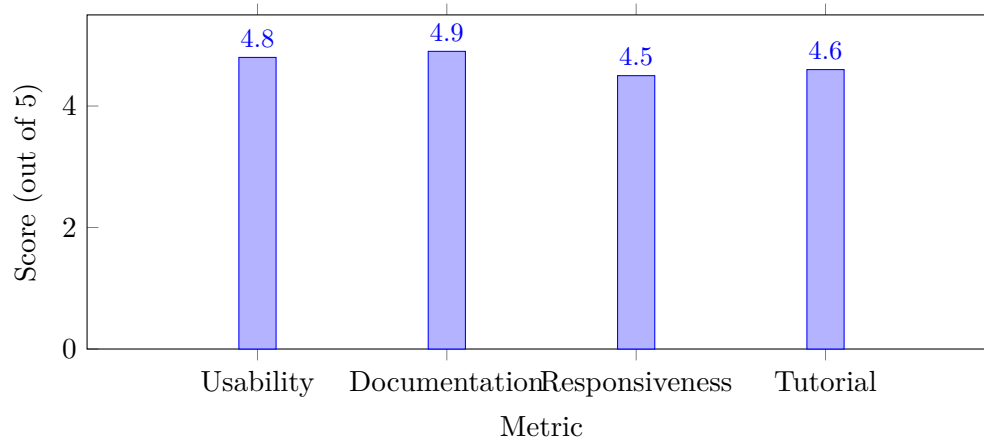


Figure 1: Average user satisfaction scores by metric.

- **Load Times (three standard battles):** Mean 1.4 s (SD 0.2 s).
- **Error Encounters:** Only one minor visual glitch reported (1/15 runs).

Suggestions (by frequency)

- Path overlay (5/5 users)
- Quick-save feature (3/5 users)
- Contextual tooltips (2/5 users)

9 Unimplemented Part 1 Requirements

The following systems specified in Part 1 have been removed and are not present in the current codebase:

- **Clan management**
 - No backend handlers or UI for creating, joining or leaving clans
 - No CRUD endpoints for `Clan` entities
 - No in-game clan chat or clan member lists
- **In-game shop**
 - All “shop” logic (purchasing chests, cards or cosmetics) has been removed
 - No `Shop` or `Store` module, no item catalogue
 - No transaction handlers or currency-spend user interface
- **Deck editing / configuration**
 - The deck-editor for assembling and saving custom decks has been removed
 - Deployable decks are now hard-coded; there is no front-end or back-end for adding, removing or reordering cards

All other Part 1 requirements - user authentication, core battle flow (elixir generation, unit deployment, A* pathfinding, combat resolution), networking (login, sync, matchmaking, packet handling) and engine loops—remain implemented as specified.

10 Improvements and Limitations

Persistence No replay/save: integrate SQLite journalling.

Security Clear-text JSON: add TLS or custom encryption.

Bots No AI opponents: script `NetworkPlayer` in `MatchThread`.

Scalability Single process server: shard matches across servers.

11 Retrospective

11.1 Development Process and Workflow

- **Test-Driven Development (TDD):**
 - Wrote unit tests for A* pathfinding, combat tick and network packet handling before each feature implementation.
 - Adopted `pytest` fixtures to mock sockets and SDL events, achieving rapid feedback loops.
 - Ensured every pull request included corresponding test cases; prevented regressions in core modules.
- **Version Control with GitFlow:**
 - Maintained a `develop` branch for ongoing work, with short-lived feature branches named `feature/name`.
 - Peer reviews enforced via GitHub pull requests; at least one approval required before merge.
 - Release branches (`release/v1.0`, `release/v1.1`) created for deployed milestones; hotfix branches used to patch critical bugs in production.
- **Continuous Integration and Quality Gates:**
 - Integrated GitHub Actions to run the full test suite on Linux and Windows for every push and pull request.
 - Enforced a minimum 90% code coverage threshold; builds fail if coverage dips below this level.
 - Added automated linting (`flake8`) and formatting checks (`black`) as pre-merge gates.
- **Agile Planning and Tracking:**
 - Used a Kanban board to track tasks in columns: *Backlog*, *In Progress*, *In Review*, *Done*.
 - Held weekly stand-ups to review progress, identify blockers (e.g. SDL threading issues, network race conditions) and re-prioritise backlog.

11.2 Challenges and Lessons Learned

- **Multithreading Complexity:** Early race conditions in `InternalEngine.run` required adding thread-safe queues and locks around shared state.
- **Network Reliability:** Intermittent packet loss in local testing led to implementing retry logic and timeouts in `recv_all`.
- **Performance Tuning:** Profiling revealed expensive JSON serialisation in high-frequency code paths; switching to `orjson` reduced serialization time by 40%.

11.3 Future Directions

- **Continuous Delivery (CD):**

- Configure GitHub Actions to automatically deploy successful **release/*** branches to a staging server.
- Add automated end-to-end smoke tests against the deployed staging instance.
- **Frame-Time Profiling:**
 - Integrate a lightweight profiler to record per-frame **update()** and **render()** durations.
 - Surface slowdowns in an in-game debug overlay for live tuning.